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Use of GPS Tracking to Interactively Explore Disabled Children's Mobility and Accessibility Patterns

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Abstract

This paper presents a methodology that addresses the need to assess disabled children's mobility and access in the context of the built environment. A novel combination of automated Global Positioning System tracking technologies, visualization using time diaries and 3-D maps, and interactive interviewing with disabled children is used to explore the real-world accessibility issues these children face. Pilot test results are used to illustrate the multi-stage method of data collection, a summary of results, and methodological challenges. Future studies should capitalize on automated activity tracking to minimize the need for participant recall and maximize the time spent engaging participants in discussion of pertinent issues.

Keywords: disabled children, mobility, accessibility, global positioning systems, activity tracking

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Background

Since the United Nations passed the Convention on the Rights of the Child in 1990 there has been an explosion of interest in the link between children's social participation and accessibility and their well-being (Morris 2001; Beckung and Hagbert 2002; Hammal, Jarvis and Colver 2004; King et al. 2006; Heah et al. 2007; Law et al. 2007; Majnemer et al. 2008). Physically disabled children's mobility and accessibility have a profound impact on their participation and inclusion and therefore on their health and well-being. The extent to which they participate inclusively affects their activities, social positioning and sense of belonging (Baker and Donelly 2001; Hemmingson and Borell 2002; Holt 2003; Palisano et al. 2003; Imrie 2004; Lawlor et al. 2006; Law et al. 2007). Furthermore, accumulating evidence indicates that experiences of marginalization and exclusion have negative physical and psychological health consequences (de Winter, Baerveldt and Kooistra 1999; Eisenberger, Lieberman and Williams 2003; MacDonald and Leary 2005).

It is widely agreed that the nature and extent of children's participation in activities and accessibility to places are strongly influenced by their everyday environments (Law et al. 2007). However, relatively little research has focused on identifying and describing environmental barriers to participation for children with disabilities (Alriksson-Schmidt, Wallander and Biasini 2007) and little is known about disabled children's leisure activities and preferences (King et al. 2006). There is also limited knowledge about how children's mobility changes in various environments and with mobility methods used (Palisano et al. 2003). The research that has been undertaken underlines the centrality of geographical features of environments to disabled children's experiences of inclusion.

Measuring Mobility and Activity Participation

Many instruments have been developed to assess mobility in disabled children, including the Gross Motor Function Classification System, the Pediatric Evaluation of Disability Inventory, the Activities Scale for Kids, the Gillette Functional Assessment Questionnaire walking scale and the Functional Mobility Scale (see review in Ravesteyn et al. 2009). These instruments assess mobility and activity limitations in terms of motor functions (Palisano et al. 2003) and have contributed significantly to the improved prediction of restrictions in disabled children's education and social contexts (Beckung and Hagbert 2002).

Although to some extent these measures predict children's ability to participate socially, the measures remain limited by their focus on children's physical limitations, as if these could be separated from their environments and the affordances that those environments provide. Lawlor et al. (2006, 220) argue that this reflects the strong tendency for researchers to rely on "a medical model of disability in which the important influences on participation are presumed to be impairments [in the individual] rather than environmental factors."

However, more recent researchers have begun to document the important influence of physical, social and attitudinal environments on disabled children's participation, such as the district of residence, family preferences, and social supports (Majnemer

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et al. 2008). These studies have elicited qualitative and quantitative data. Some are child-administered, some are completed by parents, and others are based on retrospective interviews with children (King et al. 2006). Some researchers have focused on children's environments using direct observational techniques, but these tend to be very time- and resource-intensive (Clarke et al. 2008; Yantzi, Young and Mckeever 2010). For example, Clarke et al. (2008, 506) assessed the effects of built environments on disabled adults' mobility by "using systematic social observation to independently rate street and sidewalk quality in the block surrounding each respondent's residence." They found that for those with severe physical mobility limitations, poor conditions such as potholes and broken curbs negatively affected their mobility four times more than for those with mild or no physical disabilities. Other researchers have attempted to examine person-environment fit through small-scale intensive interviews or focus groups designed to elicit the features of environments known to restrict or facilitate participation and inclusion (e.g., Hemmingson and Borell 2002; Heah et al. 2007).

One of the most comprehensive quantitative methods for assessing how disabled children's participation is altered by different environments is the Children's Assessment of Participation and Enjoyment (CAPE; King et al. 2006). CAPE measures children's self-reported participation and enjoyment, including both formal (classes, sports teams, lessons) and informal (reading, talking on the phone) activities. Children report what activities they participate in and how often, and researchers follow up with them to determine with whom they do these activities, where they do them, and how much they enjoy doing them (Law et al. 2006). Doing activities with others is as important as the types of activities that are undertaken; understanding the social context within which children are participating and with whom is critical to designing and implementing interventions (King et al. 2006). Additionally, CAPE has been paired with descriptive statistics, interviews and quantitative function assessments to identify the mix of factors that influence participation (Majnemer et al. 2008; Egilson and Traustadottir 2009).

Review of Methodological Approaches

Despite the range of methods used to study the mobility, accessibility and participation of disabled children, methods are needed to simultaneously study the intersection of physical environments and children. Pearce, Witten and Bartie (2006, 389) note that "there has been a scarcity of studies that have directly measured the material characteristics of neighborhoods theorized to have an impact on health and health inequalities" and argue that the consequences of this lack of knowledge is widespread and extends beyond disabled people. Since Pearce and colleagues' call, there have been a number of studies that have examined the relationship between neighborhood qualities and health (see Warr et al. 2007; Wilson et al. 2009; Wilson et al. 2010; Carroll-Scott et al. 2013). Yet, there remains a gap in studies specifically examining the quality of neighborhood environments in which disabled children live. Until the intersections of people and places are understood, critical information needed to improve policies and built environments will be missing. Although such interventions are important to all people, they are vital for enabling children's participation and inclusion in community environments (Majnemer et al. 2008).

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Recently, researchers have used Global Positioning Systems (GPS) and Geographic Information Systems (GIS) as part of a suite of data collection tools to evaluate the impact of environmental attributes on the physical activity, health and school travel patterns of non-disabled children (see Mackett et al. 2007; Wiehe et al. 2008; Duncan, Badland and Schofield 2009; Cooper et al. 2010; Maddison et al. 2010). Additionally, researchers have used such technologies to examine the differences in child activity patterns when an adult is present or absent (Mackett et al. 2007), during weekdays and on weekends (Maddison et al. 2010), and in indoor and outdoor environments (Cooper et al. 2010).

However, where there has been an increase in accessing children to participate in research, it is uncommon for disabled children to participate directly (Garth and Aroni 2003). Hammal, Jarvis and Colver (2004) conducted a study to assess whether children's self-report was reliable and/or necessary and found that while parents and children agreed in a number of areas, data about children's emotional function had the lowest concordance. They concluded that it is important to determine children's own perspectives. This is particularly important when exploring feelings of belonging or inclusion because these perspectives often differ between children and parents (Garth and Aroni 2003; Davis et al. 2007). Involving disabled children in the research process is thus increasingly accepted and even expected (Nordström 2010).

Data elicitation techniques for children have also undergone significant development in the last ten years. The use of cues, props, non-verbal communication strategies like drawing, and techniques such as mapping time lines, calendars, diaries and photos increasingly are acknowledged (Morrow 2001; Young and Barrett 2001; Cappello 2005). There is some evidence that children as young as 3-5 years can recognize and interpret maps, even before they have been exposed to or taught to read them (Blaut, Stea and Spencer 2003). Cues, props and non-verbal forms of communication can improve the ability of children to recall and retrieve information, especially cues that are sensory rather than semantic or textual (Salmon 2001). Most children are familiar and comfortable with drawings, games and exercises, while some find it hard to convey their feelings and opinions readily in words to adults. In that children are also motivated by things that are interesting and fun, they seem to be attracted to methods that are pleasurable (King et al. 2006). Boredom can plague research with children, but this can be overcome through formats that engage and sustain their interest (Hill, Laybourn and Borland 1996).

With these issues in mind, in the study reported here we used GPS tracking technology translated into visual data in combination with interviews of children. This design meets many of the requirements for reliable, consistent methods that directly assess children's environments and listen closely to children themselves. The combination of GPS tracking (overlaid on maps of the built environment created with a GIS) with detailed discussions with children explores the person-environment fit that is at the heart of mobility/participation concerns. Researchers have used innovative and creative ways of combining GIS information with

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qualitative data (e.g. Matthews, Detwiler and Burton 2005), but few have taken the next step to visualize the data, show them to participants and engage them in conversation about them. As Rainham et al. (2010, 674) state, "methods and technologies for time geographical research are currently available, and should be exploited more fully in empirical research efforts." We posited that using both GPS technology and child-friendly discussion would provide the precision of GPS data, supported by the rich details gained through and participatory qualitative interviews (Matthews, Detwiler and Burton 2005; Elgethun et al. 2007). Using automated GPS tracking also relieves some of the burden of participants' recollection and allows one to capture data that represents actual events that can subsequently be elaborated on by participants (Doherty 2009; Doherty and Oh 2012).

Objectives

This paper contributes to the exploration of disabled children's mobility patterns and accessibility issues in their environments. Methodologically, it presents a novel combination of the use of automated tracking technologies, visualization using time diaries and 3-D GIS maps, and interactive interviewing with children to provide comprehensive and accurate observation under real-world conditions. This approach directly addresses the need to develop effective methods to assess and engage with both environments and children directly. With respect to empirical results, this paper is suggestive rather than conclusive concerning the mobility patterns and accessibility issues discovered using this approach. Overall, this paper demonstrates the potential of several emerging geographic technologies to the study of public health issues.

Methods

The methods and results reported in this paper stem from a collaborative interdisciplinary study entitled "School, Homes and Neighbourhood Accessibility (The ScHaN Project)." It focused broadly on determining the accessibility and quality of disabled children's homes, schools and neighborhoods, specifically how these children navigate and evaluate these three built environments. Staff at Children's Treatment Centres recruited study participants across rural and urban regions of Ontario, Canada. Criteria for participation included: being aged 9 to 14, using at least one mobility device at least part-time, attending an elementary school, communicating in English, and having a parent/guardian willing to support participation.[i]

Thirteen children were recruited, including two initial pilot participants that helped refine data elicitation techniques. The final 11 participants represented a diversity of mobility-disabled children and had the following characteristics:

- four males, seven females
- ages 10 to 14, median/mode 12
- two lived with mother only, nine live with two parents/guardians
- five lived in urban areas and six lived in rural
- eight had Cerebral Palsy, two Muscular Dystrophy, and one Spina Bifida

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- two used one mobility device, nine used two or more mobility devices, including manual wheelchair (7), electric wheelchair (5), braces/splints (5), walkers (4), crutches/canes (2), and orthotics (2)

Gift baskets were given to each family on the final day of data collection in appreciation for their time and participation. In the first meeting with each participating child and parent(s), research assistants (RAs) described the case study series that included three face-to-face home interviews and a two-day period of collecting the child's neighborhood and community mobility patterns through a GPS enabled smartphone carried by the participating child. RAs asked participants to take the smartphone whenever they left their homes and to leave it turned on if possible on at least one school day and one weekend day. A clip-on hardcover holster was provided to protect the device and keep it in sleep mode to conserve power. Children were given the option to secure the smartphone in the clip-on holster to their clothing or mobility device or to place it in a small pouch provided, which they could attach to their mobility device. No other interaction with the device was required, other than asking the children and a parent to plug in a provided cable at the end of the day to recharge batteries. RAs informed participants that the GPS data would be used to generate time diaries and Google Earth maps, which would be reviewed with them in a follow-up interview using a laptop computer.

As the contribution of this paper is primarily methodological, and because the various methods involve fairly experimental technologies and interview techniques, significant space is devoted to describing each stage in the study, as follows.

Stage 1a: Remote Monitoring

Remote monitoring of children's two-day activity and mobility patterns was accomplished by using GPS-enabled smartphones outfitted with custom software that transmitted data to a remote server for storage and subsequent processing. The smartphone was a BlackBerry 8330 with an embedded Sirf Star III GPS receiver, capable of providing a second-by-second location to within about 5 meters accuracy in NMEA (National Marine Electronics Association) standard format. This includes longitude, latitude, time, speed, and signal quality. Custom Java (micro edition) software was developed expressly for the BlackBerry to run continuously in the background, logging the GPS data, compressing it, and sending it in small packets wirelessly (about every five minutes) to a central server.

A Blackberry Enterprise Server was set up to enhance security of the data and to further allow the smartphone to be remotely monitored while "in the field." Most importantly, this allowed the status of the GPS receiver, device, and on-board software to be monitored by RAs to assure that the smartphones were working properly at all times. This helped to avoid and/or quickly address equipment issues such as program freezes, low battery, and loss of GPS signal that can occur. In cases where they did occur, automated messages were sent to research staff to alert them to subsequently access an in-house on-line web application to view all details on the status of the device and "push" commands to the device to solve the issue (e.g. Java program restarts, device restart). As a last resort they contacted

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the participants/parents to solve the problem. These efforts helped increase the data capture rates to over 80 percent.

Stage 1b: Establish Preliminary Diary

Wireless transmission of the GPS data to a central server allowed the data to be viewed and processed in a timely fashion to automatically determine participants' mobility and activity patterns. When manually viewing GPS tracked data on a map, a wide range of activities and trips became immediately apparent to the trained RAs, including start/end times, locations (including buildings names if data was available), travel routes, and even travel mode on careful inspection. Less apparent were the specific types of activities performed at locations where participants were stationary or in small areas, other persons present, and of course, the real-time thoughts and feelings of participants.

To arrive at an accurate and complete "diary" for each child's daily mobility patterns, a combination of automated and manual techniques was first developed "in the lab." An automated algorithm was created to initially process the raw un-cleaned GPS data to identify episodes of travel versus episodes of stationary activities, including start/end times, likely travel mode, and location. For instance, travel mode was detected using a set of if-then rules that utilized a wide range of explanatory variables calculated at one second intervals, including speed, acceleration and contextual information such as distance to nearest roadway (e.g., if speed > x km/hr, and distance to roadway < y meters, then mode is z). Rules for detecting stationary activities were based on identification of a sufficient number of GPS points within close proximity of each other. After determining the geo-coordinates of stationary activities, other rules were used to determine attributes of the location, such as if it was located within a building (using building footprint data if available). Lengthy episodes depicting missing GPS data were also identified. Most were the result of lost GPS signals when children entered an indoor location.

The results of the automated algorithm was a table of events for each child that could subsequently be displayed on screen using a custom made web-based diary programmed using Java (Figure 1a). Five columns in the diary corresponded to event attributes including: *date*; start/end *time* of event; *event type* (activity or trip), followed by drop-down lists of sub-categories (e.g. Activity: sleeping, shopping; Trip: automobile, bus); *location* to verify location via an interactive map; and *people* to note other persons present at the event. At first, event types were limited to either activity or trip, locations were displayed on an interactive map by centroid with a label derived from a local search (e.g. "Residential location near King Street"), and the column for people was left blank. All attributes could then be interactively modified and/or updated to be more specific.

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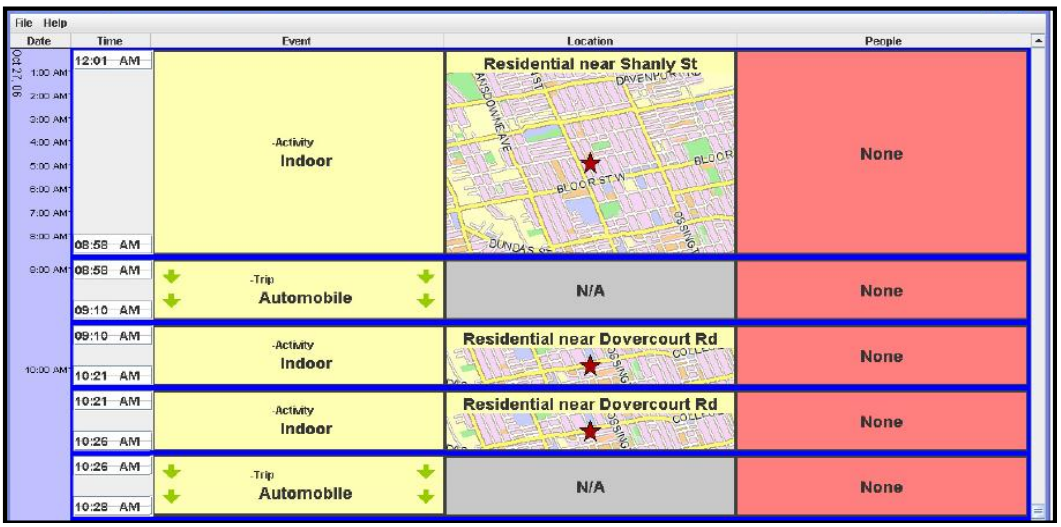
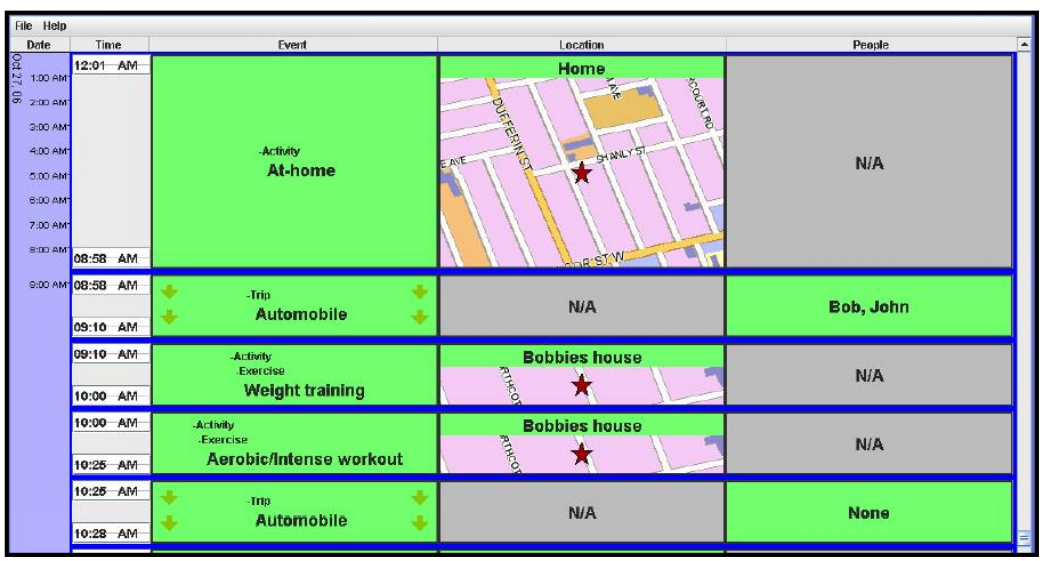
Figure 1. On-line interactive activity-travel diary showing example data

a) Raw output of automated detection algorithm



b) After manual inspection and interaction with participants



Notes:

- The diary is interactive, and users click on attributes on the screen to update/modify them.
- Red indicates wholly unknown attributes, yellow indicates attributes that require confirmation or updating with more detail, and green indicates attributes that are confirmed and completed.

Prior to the follow-up home interview with each participant to discuss the mobility data, RAs manually inspected each participant's diary to refine it and correct missing GPS data. The RAs were instructed to only make adjustments that were

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obvious or that were certain based on available evidence (RAs could also view raw GPS data in a GIS for further clues). This refining consisted of the following tasks:

1) Converting episodes of missing GPS data into activities: this typically involved lengthening a previous stationary activity up to the start of the next trip. This was usually obvious on viewing of the GPS data, as the trip to the activity ended in a small cloud of GPS points at the location, a period(s) of missing GPS points, followed by a re-acquired GPS-tracked trip leaving the location.
2) Correcting trip start times owing to GPS warm up time: this typically involved imputing 30-90 seconds of missing trip time.
3) Refining location labels when possible: "Home" and "School" labels were added, as these were known mobility destinations.
4) Omitting short activities that were obviously traffic light stops.
5) Omitting very short "trips" (< 1 minute) within stationary activities that were more obviously short movements indoors or wildly scattered GPS points indicative of signal error.

These labor/time-intensive efforts were conducted to reduce burden, but not eliminate children's involvement in providing accurate and complete diaries. This work went a long way towards addressing the well-known difficulties people have recalling start/end times accurately, and specifying locations succinctly (e.g., street addresses). However, because GPS could not provide all the diary details required for this study, and because verification of the accuracy of automatically detected events was desired, a manual review of the diary and prompting for refined/additional attributes was conducted subsequently with each participant.

Stage 1c: Interactive Diary Review and Establishing Final Diary

A laptop computer with roaming wireless Internet capabilities was used in the follow-up home interview conducted to further review and refine the on-line diary with each child. This interview was conducted within five days of the GPS tracking period. RAs performed all data entry, completing the following tasks:

1) Confirming the accuracy of automatically detected and imputed events from Stage 1b.
2) Filling in unknown activity time periods (not previously imputed in the lab).
3) Providing more detailed event types using fly-out lists.
4) Adding detailed location names (e.g. "John's House," instead of "Residence near King St.").
5) Adding other persons present.

Stage 2: Generation of Combined Spatial-Temporal Activity Data for Viewing in Google Earth

To this stage, an unprecedented amount of spatial and temporal detail on participants' movement from place to place, activity and mobility had been collected, involving second-by-second data. The challenge was how to best display this data in a recognizable way without overwhelming the children, yet retaining critical detail for discussion that ultimately highlighted potential mobility or

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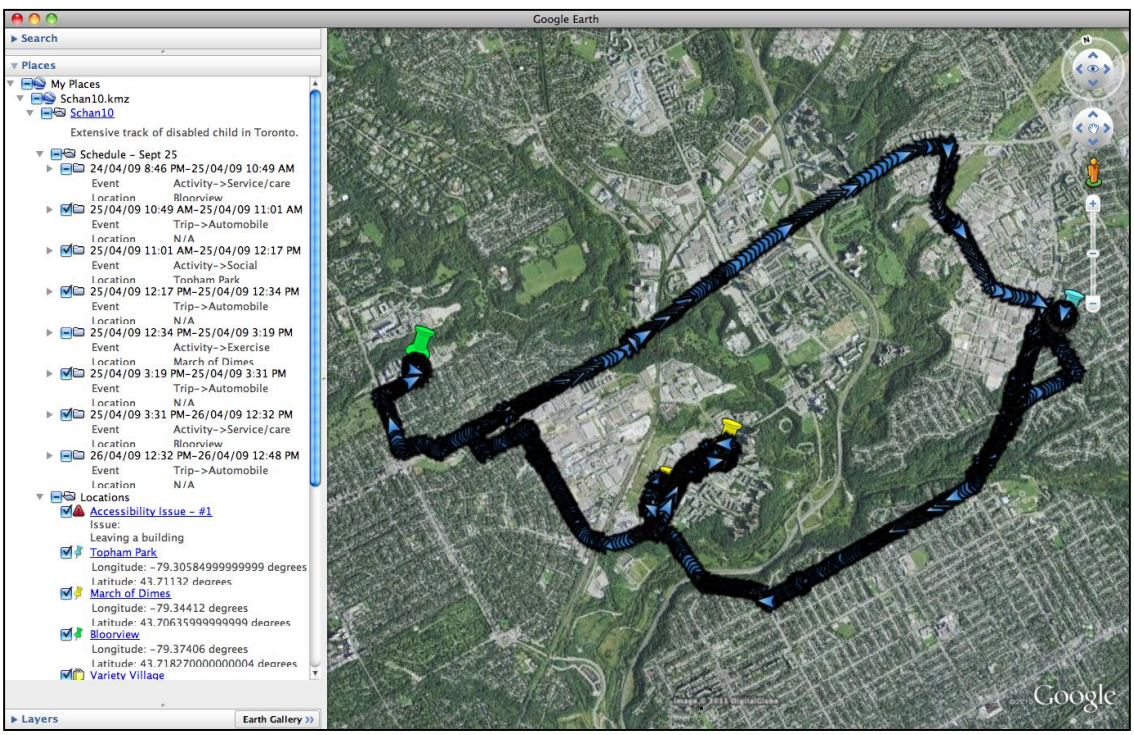
accessibility issues experienced with maximum precision and contextual cues. This would allow RAs to engage participants in in-depth discussions.

We chose Google Earth for this purpose owing to its ease of use, 3-D capabilities, variety of available base layer data, and cost-free availability. Google Earth is an on-line GIS that presents a virtual 3-D display of the globe via superimposition of a variety of spatial data including satellite images, aerial photos, street-level photos, digital elevation, GPS track and waypoint data, and 3-D buildings and structures. Panning, zooming, oblique angle viewing, and street-level viewing allow unique viewing capabilities.

The GPS-tracked data gathered in Stage 1a and the diary data from Stage 1c were combined into an interactive KML (Keyhole Markup Language) file that was viewable in Google Earth (see Figure 2 for an example). File generations were accomplished via a custom-made server-based program that RAs used to generate the files while logged into the secure server. The files contained GPS-tracked point data sorted into folders (see left side of Figure 2) that represent each of the events included in the diary, along with waypoints representing all visited locations.

Figure 2. Example space-time path snapshot in Google Earth™



Notes:

- See Figure 3 for an example zoomed-in snapshot of individual GPS points.
- The "Schedule– Sept 25" timeline at left shows start-end times and locations of individual trips and activities, organized into folders that can be checked on-off for display.
- The "Locations" folder lists all destinations (labels provided by participants) and the location of accessibility issues discussed in interview.
- Some locations have been modified to protect the identity of participants.

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The following principles were applied to displaying the GPS data to make spatial-temporal human activity patterns/paths more easily interpretable by the naked eye:

1. Layering: Overlaying different layers of data, such as road networks, land-use, and aerial photos
2. Connect-the-dots: Connecting GPS points with a line (especially useful when paths overlap)
3. Directionality: Conveying the heading/direction of each GPS point with a directional icon or arrow
4. Speed: Shade or color GPS icons according to speed; stationary activities with near-zero speeds stood out as clouds of distinct color
5. Time: Displaying the time beside occasional GPS points (care was taken so this did not become too cumbersome visually)
6. Spatial Means: Large sets of GPS points at stationary locations were replaced with a single icon representing the spatial mean of all points; this was useful as GPS points tended to "wander" even when perfectly stationary, resulting in large random-looking clouds of points at activity locations that could obscure the view and increase file sizes.

As a final preparatory task, especially for publishable images, it was necessary to maintain the children's anonymity. This was a particular challenge with such data because home or school locations were easily identifiable on close inspection. The most direct approach would be to leave out such locations from any images. Saving that, the goal was to sufficiently "mask" the data such that patterns are still discernable while children's identities clearly are not. Armstrong, Rushton and Zimmerman (1999) propose several categories of geographic record and/or attribute "masks" that can be utilized, including "Record-transforming" and "Attribute-transforming" masks that involve aggregating similar records/attributes, suppression of certain records/attributes, sampling of records, and/or multiplication or displacement of records/attributes by random noise. In the case of this data, the calculation of "spatial means" of stationary locations served as a form of aggregation mask (i.e., aggregation of all points into a single point), but what was added was a displacement of this single point by a random amount sufficient to mask the actual location of the activity (at least 100m random displacement, although the actual amounts are not disclosed). Nearby travel points into and away from the location also required suppression or adjustment. In low-density areas, the random amount was increased even more.

Stage 3: In-Depth Discussion of Accessibility Issues

The ultimate use of the preliminary diaries described above was to serve as a basis for in-depth discussion of actual accessibility issues faced by the children. While preparing the data in the preceding stages, RAs took careful notes on any unusual situations that appeared to involve an accessibility problem or issue, such as very slow trips, intermittent, unusual or complex movement patterns, stationary activities in unusual places, long stops, or any other challenging situation. These were used as prompts during discussion with participants.

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A "conversational" interview approach was adopted with each child. The conversation started with the statement, "The first thing we'll do today is talk about the map the BlackBerry made while you were wearing it. I hope you find it interesting to see your map on the computer. Let's talk about the places you went." During this conversation, participants were asked more specific questions including, "Did you use any mobility devices?" "Did anyone help you?" "Did you have any problem entering the location… getting around in the location?" Included in this conversation were questions concerning accessibility issues that had already been detected by the RA, in the form of, "Can you tell me what happened here…?" Throughout, the Google Earth display was consulted and explored extensively, including zooming in on certain activities, movement patterns or situations (including Streetviews, which are ground level pictures that move as you navigate), in an effort to jog memories and flesh out the details of the potential accessibility issues. The computer screen activities and verbal discussions were recorded using Camtasia Studios screen-capture software in support of later analysis. This software is analogous to pointing a video camera at the screen and recording sound.

All accessibility issues that were closely associated with a location in space were added to Google Earth space-time paths as location icons. When clicked on screen, a detailed description of the issue popped up in a call-out box.

Results

Of the 11 child participants, one forgot to carry the device outside the home, leaving ten participants. These ten participants provided complete two-day GPS-assisted diaries. In two of these cases, the days were not sequential owing to equipment problems that occurred after the first day (in one, the smartphone ceased working, and in the other, data were lost in transmission). Both participants agreed to carry their smartphones on the subsequent day. Initially the diaries included start/end times of activities' locations, and other persons present.

On average, participants spent six hours engaged in activities outside the home, including school, leisure/recreation events (e.g., movies, museums, pools, visiting), and other activities (e.g., restaurants, shopping). These consisted of an average of four one-way trips per day with an average length of 24 minutes each. In almost three-quarters of these trips (72 percent), the means of transportation was a car, 8 percent were by bus, and 20 percent involved walking or wheelchair. Remaining time was spent in their home.

Spatial-Temporal Visualization of Mobility Patterns

The GPS data mapping principles laid out above in the Methods section aimed to enhance the visualization and interpretation of human activity-travel patterns at varying spatial-temporal scales along with the situation and environment in which they occurred. Figures 2 through 5 illustrate the results. As shown in several of the close-ups in the figures, the interactive nature of Google Earth easily allowed zooming/panning into specific situations to explore events more thoroughly. Note especially how closely the GPS data matched the road network, how repeated

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tracks on the same roadway are distinctive, how intersection navigation is evident, and how a wide range of specific behaviors are evident such as parking searches, building entrances and indoor movements, etc.

One of the more efficient ways to view the data is to display only GPS data associated with trips, along with a single waypoint icon at the spatial median of activity locations (adjusted to mask true locations). This effectively replaces the large distracting clouds of GPS points at locations. This was done very simply in the Google Earth file by selecting the desired trip and location folders in the "Places" pane at left, as shown in Figure 2. In the same location, notice also how traditional activity/travel data from the diary is integrated into Google Earth in the "Schedule" folder, enabling the selection of diary items for display and exploring associated activity types, location names, and start-end times.

None of the presented figures fully captures the capabilities that are revealed by experimenting live with the various zooming/panning and layering of different information. Interested readers can download examples of Google Earth kmz files at the first author's homepage (see http://www.wlu.ca/~sdoherty).

This approach and these results build strategically on traditional space-time maps derived from diary data that typically are limited to displaying locations linked by straight lines, as pioneered by Hägerstrand (1970), and displayed commonly using GIS. However several areas for improvement remain, including dealing with the scattering of GPS points at stationary locations, increased error of GPS points near large buildings, and some small gaps in GPS data. Additionally, for two of the 11 participants, data views like those described above were not possible because the participants lived and engaged in activities in remote rural areas where Google Earth images were not clear enough to enable clear identification of roadways and buildings.

Accessibility Issues

The purpose of collecting and preparing the data for viewing in Google Earth was to provide a basis for in-depth discussion of actual real-world mobility and accessibility issues disabled children had encountered in built environments. Of the 11 participants, eight were engaged in in-depth discussions in this manner. For two participants, the Google Earth displays were not clear enough for discussion (noted above) and one child could not be reached for a follow-up interview.

The eight participants discussed 15 significant accessibility issues faced during their two GPS tracking days—equivalent to about one issue per day. All but one barrier had been identified prior to the discussion by RAs. Examples are provided in Table 1 and associated Google Earth snapshot images are provided in Figures 3 through 5. These examples illustrate the methodology and typical issues and are not comprehensive. As illustrated in Table 1, participants faced a variety of accessibility issues in their neighborhoods and communities. Most frequently these resulted in children waiting in vehicles or outside of buildings for extended periods in lieu of participating in activities (i.e., sitting in the car while a parent went into stores). They described difficulties they experienced getting in/out and around

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buildings, making slow intermittent/wandering trips in awkward outdoor places, and taking long multi-stop rides with significant wait times, especially on bus rides to and from schools. These issues occurred around their homes, schools, on roadways, at retail stores, and in recreational areas. Accessibility issues occurred when children were with family members, friends and caregivers. Anecdotally, many participants said they did not recall the length of time they had waited, or the number of stops they had to make and expressed surprise by the length of the tracked values in Google Earth. Participants also voiced a considerable degree of complacency and resignation with many of the issues and barriers they faced.

Table 1. Example accessibility issues identified and discussed during in-depth interviews

Issues	Location	Research Assistant Observation Prior to Discussion	Participant Notes/Explanation During In-depth Discussion
Left in Car	Store parking lot (Figure 3)	Subject appeared to do an activity in a store parking lot or was left in the car (or left the device in the car)	Subject and her sister stayed in the car; Mom says it's faster to shop that way
Getting out of Car and Walking to Beach	Beach (Figure 4)	Appears to be a clump of points by the parking lot then a very slow walk over to the beach; the return trip looks similar	Subject explains that it takes a while to get out of the backseat because it's a two-door car. Subject doesn't use his walking aid outside of school, and ends up falling a lot while walking.
Leaving a Building	Home	Subject appeared outside the home for over 5 minutes	Subject explained that some days it takes longer to leave the house.
Long Wandering Trip	Home to movies (Figure 5)	When zooming in on 30-minute trip, participant appeared to wander all over the place rather than in a straight line	Subject explained that the lack of sidewalks meant she often traversed on the road but has to move away for cars and frequently rolls to the sides because of road slopes.
Long Bus Route	On route to school	The route pattern shows that it took 45 minutes to arrive at school and that multiple stops were made	The participant thought her bus ride was only 15 minutes and made one stop. After seeing her route she remembers the other stops.
Left Outside	School	Subject did not appear to go into the school building.	Subject was told to wait outside for the rest of the class, because he takes too long to walk in and out and it would be faster just waiting outside alone.
Entering a Building	Large shopping mall	Subject appeared to have some issue entering the mall, as it took 5 minutes.	Subject explained that he faced a physical barrier because the automatic doors did not work, so it took longer to enter the building.
Waiting for the Bus	Home	Subject appeared to be outside her home for 15 minutes in the morning.	Subject explained it takes 10 minutes to get out of her house, 5 minutes to wait for her bus, and 10 minutes to load the bus every day.

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Figure 3. Example accessibility issues: Being left alone in parking lot

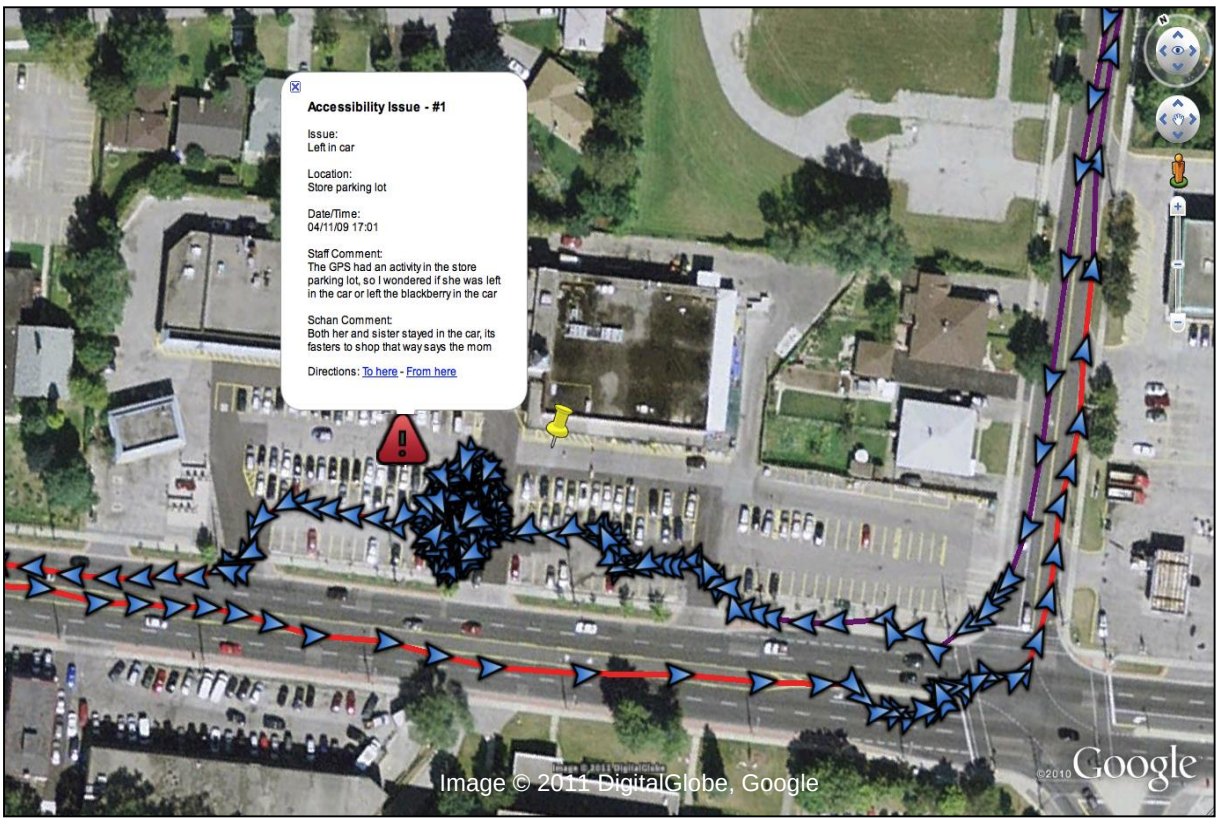

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Figure 4. Example accessibility issues: Difficulty getting to the beach

Figure 4a. Enlarged section

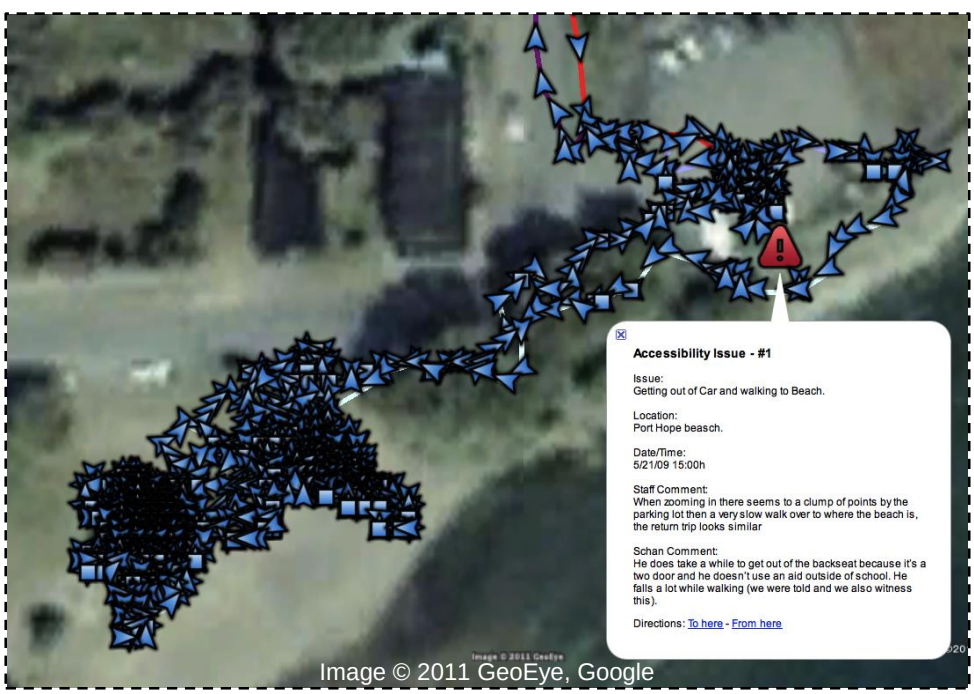

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Figure 5. Example accessibility issues: Meandering trip along roadway without sidewalks

Figure 5a. Google Streetview depicting participant navigating around parked cars as no sidewalks exist

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Combining accessibility issues with the GPS-tracked activity-travel patterns into comprehensive displays was accomplished by overlaying locations and providing the qualitative details of the issue elicited during the interview as text attributes of locations. A special yield-sign icon was used to highlight these locations. Clicking on the marker on the map yielded a pop-up text box explaining the issue, as shown in Figures 3 and 4. This approach provided valuable qualitative information on issues that GPS-based patterns alone cannot reveal, and allowed for a much greater appreciation of the site and situation.

Additional dramatic views and interpretation of the data were possible by turning on 3-D buildings or going into Streetview. For instance, in Figure 5, it was possible to closely examine the route the participant took, providing considerable elaboration into the difficulties/hazards the participant voiced during the interview when faced with no sidewalks and frequent parked cars that got in the way. It was as if the interviewer re-created the scenario from the participant's viewpoint, and thus gained a much more intimate appreciation of his navigation and mobility difficulties.

Discussion and Conclusions

This paper presented methods that address the need to assess disabled children's mobility in real-world built environments by engaging children directly in discussion and utilizing a novel combination of tracking and spatial-temporal visualization techniques. In comparison to previously published evidence, this approach enhanced the examination of child-environment fit that is deemed central to mobility, participation and well-being. It also provided the critical information necessary to make interventions in policies and in built forms. The presentation of this technique and results highlight the multi-stage nature of the methodology that involves the careful processing and display of tracked data in computer diaries. This effort allowed us to engage disabled children in meaningful discussion on accessibility issues they faced. Results clearly demonstrate the value of the features of GPS tracking (automated, accurate, real-world tracking) and how this subsequently allowed accessibility issues to be explored in more depth without burdening subjects with tedious and difficult to recall situational information. As a practical tool, this type of methodology could be used in the future as an advocacy tool by the disabled, as well as other at risk populations such as the elderly, children with chronic illnesses, or rural residents. For example, disabled people could actually show the barriers they experience to their local government officials. Geo-visualizations plus interviews could provide other types of data for use in advocating for full participation and adherence to accessibility legislation.

Several technical and operational challenges emerged that are worth further discussion. First, GPS data alone are not sufficient to capture all activities and their attributes. GPS signals frequently deteriorate (especially in buildings) leaving gaps in data, and key attributes such as specific activity types, location names and other persons present are not identifiable. To overcome this, before meeting with participants, RAs manually corrected common issues and added known location names. Participants themselves were consulted via a prompted recall diary to continue this process, and add information about involved persons during the follow up interviews. Although necessary, this was time consuming.

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The next technical challenge concerned how to display detailed GPS data back to participants, along with contextual cues from their diary, so that activities and trips were easily recognizable to participants without overwhelming them. Publically available GIS software (i.e., Google Earth) was more than sufficient for this task, however this required careful preparation of kmz files, including incorporating lines and arrows in the GPS data, carefully managing locations (including replacing large clusters of GPS data with pinpoints), and using folders to manage temporal information. Although Google Earth is freely available, the processing required some custom programming and use of (previously calibrated) custom data processing algorithms.

The results revealed a few additional technical challenges, but clearly illustrate the nature and extent of data that were elicited and insights that were gathered. Most novel, the temporal and spatial displays via Google Earth speak for themselves. Fine accuracy of the data and contextual/environmental detail were accessed including aerial, 3-D and street-level views. Ultimately, this assisted participating children to identify accessibility issues in their environmental contexts. The environmental attributes (locations, durations, involved persons) were quantified and provided a forum for further explanation by participants in their own words. Even with the small sample, considerable insights were gained into the unique types of accessibility issues faced, all but one of which were evident in the data prior to discussion with participants.

In the future, the types and amounts of data that can be tracked from human participants will continue to expand aided by the rapid proliferation of smartphones and mobile biophysical sensing devices. The true value of this data will not be realized without effective processing, analysis, and engagement with the people who are involved, as exemplified in this paper. Future large-scale studies should continue to capitalize on advancements in automated activity detection algorithms that reduce the need for manual interaction with the data, and maximize the time spent interacting with participants.

Endnote

i. The Canadian Institutes of Health Research and the Bloorview Kids Foundation Chair in Childhood Disability Studies funded this study. Ethical approval was received from the Research ethics boards at Bloorview Research Institute, Wilfrid Laurier University and Laurentian University

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visitors, and immigrants, and for the promotion of physical activity and local food consumption.

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